

**SEQUENTIAL HYDRAULIC CIRCUIT-
DESIGN AND ITS DEVELOPMENT IN
FLUIDSIM-H SOFTWARE**

By
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ABSTRACT

Fluid power in automation is the use of fluids under pressure to generate, control, and transmit power. Fluid power is subdivided into hydraulics, which uses a liquid such as mineral oil or water, and pneumatics, which uses a gas such as air or other gases. Here, in this mini-project for a minor in mechanical engineering, we deal with hydraulics because hydraulics is far better than pneumatics when dealing with strengths.

Hydraulics is a mechanical function that operates through the force of liquid pressure. In hydraulics based systems, mechanical movement is produced by contained, pumped liquid, typically through hydraulic cylinders moving pistons. Sequencing circuits are those that automatically programme two or more cylinders on a machine to cause them to extend and retract in a predetermined sequential order on every cycle. Sequence valves can improve the operation, efficiency, performance, and safety of fluid power circuits in which they are used. Sequence valves are normally closed and usually allow bidirectional flow when equipped with a bypass check valve. Sequence valves always have an external drain connected directly to the tank.

FluidSIM is a comprehensive software for the creation, simulation, instruction, and study of electropneumatic, electrohydraulic, and digital circuits. The user will be able to create circuits easily by using the drag and drop option. The user only has to place the items in the circuit as they want to view them and build the circuit intuitively. Using cascade and the KV map method, we are designing and developing a sequential hydraulic circuit in FluidSIM-H software for the sequence A+B+B-A-. Also, we are analyzing the materials used in the development of these hydraulic circuits for their manufacturing.

Keywords: “Fluid Power”, “Pneumatics”, “Hydraulics”, “FluidSIM-H”, “Cascade method”, “KV map method”.

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LIST OF ABBREVIATIONS

PLC	PROGRAMMABLE LOGIC CONTROLLER
μP	MICROPROCESSOR
KV	KARNAUGH VEITCH
ISO	INTERNATIONAL ORGANIZATION FOR STANDARDIZATION
DCV	DIRECTIONAL CONTROL VALVE
HSS	HIGH SPEED STEEL

CHAPTER 1

INTRODUCTION

1.1 FLUID POWER AND ITS SCOPE

In the industry we use three methods for transmitting power from one point to another. Mechanical transmission is through shafts, gears, chains, and belts. Electrical transmission is through wires and transformers. Fluid power is through liquids or gas in a confined space.

Fluid power is the technology that deals with the generation, control and transmission of forces and movement of mechanical element or system with the use of pressurized fluids in a confined system. Both liquids and gases are considered fluids. Fluid power system includes a hydraulic system (*hydra* meaning water in Greek) and a pneumatic system (*pneuma* meaning air in Greek). Oil hydraulic employs pressurized liquid petroleum oils and synthetic oils, and pneumatic employs compressed air that is released to the atmosphere after performing the work.

By the term “fluid” we refer to air or oil, for it has been shown that water has certain drawbacks in the transmission of hydraulic power in machine operation and control. Commercially, pure water contains various chemicals (some deliberately included) and also foreign matter, and unless special precautions are taken when it is used, it is nearly impossible to maintain valves and working surfaces in satisfactory condition. In the cases where the hydraulic system is closed (i.e., the one with a self-contained unit that serves one machine or one small group of machines), oil is commonly used, thus providing, in addition to power transmission, benefits of lubrication not afforded by water as well as increased life and efficiency of packings and valves.

Fluid power applications can be classified into two major segments:

- a) Stationary hydraulics: Stationary hydraulic systems remain firmly fixed in one position. The characteristic feature is that valves are mainly solenoid operated.

- b) Mobile hydraulics: Mobile hydraulic systems move on wheels or tracks such as a tower crane or excavator truck to operate in many different locations or while moving. The characteristic feature is that the valves are frequently manually operated.

The following are the two types of hydraulic systems:

- i. Fluid transport systems: Their sole objective is the delivery of a fluid from one location to another to accomplish some useful purpose. Examples include pumping stations for pumping water to homes, cross-country gas lines, etc.
- ii. Fluid power systems: These are designed to perform work. In fluid power systems, work is obtained by pressurized fluid acting directly on a fluid cylinder or a fluid motor. A cylinder produces a force resulting in linear motion, whereas a fluid motor produces a torque resulting in rotary motion.

1.2 CLASSIFICATION OF FLUID POWER SYSTEMS

The fluid power system can be categorized as follows:

1.2.1 Based on the Control System

- a) Open-loop system: There is no feedback in the open system and performance is based on the characteristics of the individual components of the system. The open loop system is not accurate and error can be reduced by proper calibration and control.
- b) Closed-loop system: This system uses feedback in. The output of the system is fed back to a comparator by a measuring element. The comparator compares the actual output to the desired output and gives an error signal to the control element. The error is used to change the actual output and bring it closer to the desired value. A simple closed-loop system uses servo valves and an advanced system uses digital electronics.

1.2.2 Based on the Type of Control

- a) Fluid logic control: This type of system is controlled by hydraulic oil or air. The system employs fluid logic devices such as AND, NAND, OR, NOR, etc
- b) Electrical control: This type of system is controlled by electrical devices. Four basic electrical devices are used for controlling the fluid power systems: switch, relays, timers and solenoids.
- c) Electronic control: This type of system is controlled by microelectronic devices. The electronic brain is used to control the fluid power muscles for doing work. This system uses the most advanced type of electronic hardware including programmable logic controller (PLC) or microprocessor (μ P).

Hydraulics and pneumatics have almost unlimited application in the production of goods and services in nearly all sectors of the country. Several industries are dependent on the capabilities that fluid power affords. Table 1.1 summarizes few applications of fluid power.

Table 1.1 Applications of fluid power

Sl. No.	Application	Remark
1.	Agriculture	Tractors; farm equipment such as mowers, ploughs, chemical and water sprayers, fertilizer spreaders, harvesters.
2.	Automation	Automated transfer lines, robotics.
3.	Defence	Missile-launching systems, navigation controls.
4.	Machine tools	Automated machine tools, numerically controlled machine tools.
5.	Medical	Medical equipment such as breathing assistors, heart assist devices, cardiac compression

		machines, dental drives and human patient simulator.
6.	Wood working	Tree shearers, handling huge logs, feeding clamping and saw operations.

1.3 HISTORY OF FLUID POWER

Fluid power is as old as our civilization itself. Water was used for centuries to produce power by means of water wheels and air was used to turn windmills and to propel ships. Chinese used wooden valves to control water flow through bamboo pipes in 4000 BC.

Fluid power technology actually began in 1650 with the discovery of Pascal's law. Simply stated, this law says that *pressure in a fluid at rest is transmitted undiminished equally in all directions in a confined body of fluid*. Pascal found that when he rammed a cork down into a jug completely full of wine, the bottom of the jug broke and fell out.

In 1750, Bernoulli developed his law of conservation of energy for a fluid flowing in a pipeline. Both Pascal's and Bernoulli's laws operate at the heart of all fluid power applications and are used for analytical purposes. However, it was not until the Industrial Revolution of 1850 in Great Britain that these laws were actually applied to the industry.

The first use of a large hydraulic press for foregoing work was made in 1860 by Whitworth. In the next 20 years, many attempts were made to reduce the waste and excessive maintenance costs of the original type of accumulator.

Today fluid power is used extensively in practically every branch of industry. The innovative use of modern technology such as electrohydraulic closed loops, microprocessors and improved materials for component construction continues to advance the performance of fluid power systems. The military requirements kept fluid power applications and developments going at a good pace. Aviation

and aerospace industry provided the impetus for many advances in fluid power technology.

1.4 OBJECTIVE AND SCHEME OF PROJECT

The objective of our Mini project is to design and develop a sequential hydraulic circuit of the sequence A+B+B-A- in FluidSIM software. By which each of the letters indicates the cylinders and '+' and '-' sign indicates the extraction and retraction of the cylinders respectively.

The main application of the proposed sequence is found in automatic clamping and drilling operation. The design part of the circuit is analyzed by two different method (Cascade and Karnaugh Veitch). The development and its simulation part is done in the software. We are also analyzing a comparison of the two methods and identifying the best material used in the manufacturing of the hydraulic system.

1.5 FAMILIARIZATION OF FLUIDSIM-H

FluidSIM is a comprehensive software for the creation, simulation, instruction, and study of electropneumatic, electrohydraulic, and digital circuits. The user will be able to create circuits easily by using the drag and drop option. The user only has to place the items in the circuit as they want to view them and build the circuit intuitively.

FluidSIM® is the world's leading circuit diagram design and simulation program for pneumatics, hydraulics for mechanical and electrical engineering provided by Festo, an automation company.

1.5.1 Installation of FluidSIM-H Software

Step 1: Go to the Academic Software page of FluidSIM and click the download button on the right to start the download.

Step 2: Extract all the files in the folder you just downloaded and open the file

'..HomeUseSetup.exe'.

Step 3: Choose a language and click on 'Install FluidSIM 4.5d/1.70 Hydraulics'.

Step 4: Close all open applications on your computer and click Next.

Step 5: If you have not yet installed .NET Framework on your computer, you will get a message. Choose to download and install the software. FluidSIM needs .NET Framework to work. Already have it installed? Then go straight to step 7.

Step 6: Click on Close once .NET Framework is installed.

Step 7: Choose whether to install FluidSIM for each user or only for the user who is logged in.

(fluidsimenuptodown)Step 8: Choose where the circuit diagram files should be stored.

Step 9: Click Next, Accept the license agreement and proceed to Next.

Step 10: Choose where to install the software → Click Install → FluidSIM will be installed.



Fig 1.1 About Festo FluidSIM-H
(<https://fluidsim.en.uptodown.com>)

1.5.2 Analysis of FluidSIM-H Software

Click on File → New → Then a working area for constructing the hydraulic circuits will be displayed on the screen and after designing the circuit save the

work. On right side of the working window, you can see the individual tools (hydraulic elements) for designing the system. The user can drag and drop each element to the working area and connect each ones.

After designing the circuit, the user can simulate the work by clicking the start button or by clicking F9. Similarly we have,

- Stop button → F5
- Pause button → F8

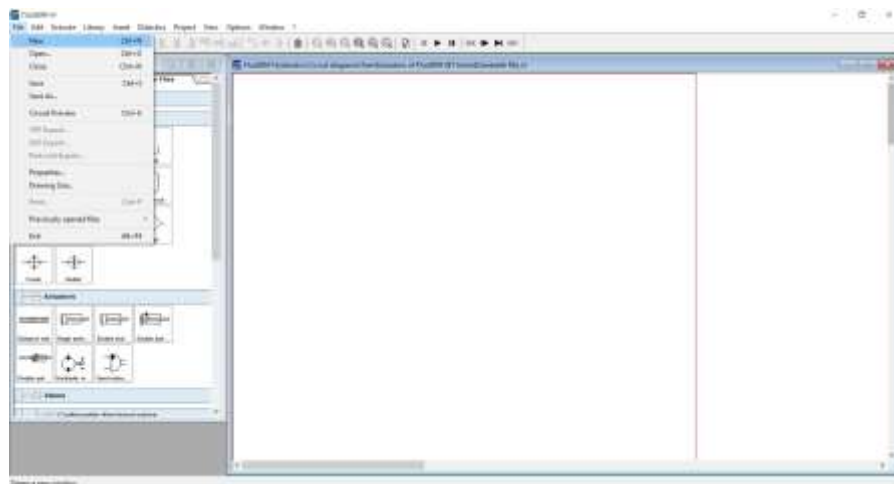


Fig 1.2 Starting a new window for designing hydraulic circuits (<https://fluidsim.en.uptodown.com>)

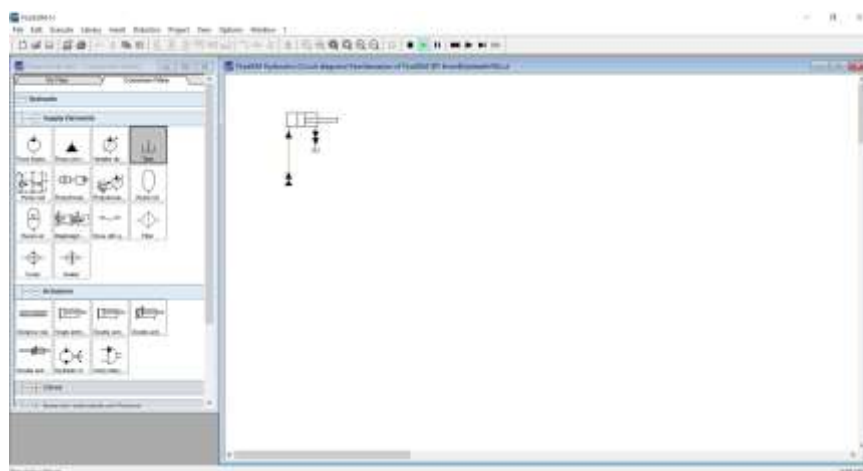


Fig 1.3 Building hydraulic circuits in FluidSIM-H (<https://fluidsim.en.uptodown.com>)

CHAPTER 2

LITERATURE REVIEW

Macro Wurster and Benjamin Hafner

Macro Wurster and Benjamin Hafner (2021) has presented their publication on Fluid Automation-A Definition and an Application in Remanufacturing Production Systems. Production systems must be able to quickly adapt to changing requirements. Especially in the field of remanufacturing, the uncertainty in the state of the incoming products is very high. Several adaptation mechanisms can be applied leading to agile and changeable production systems.

Jose Incera and Raymond Marie

Jose and Raymond (2001) had described a software for the actuation of the proposed hydraulic circuits in FluidSim software, a tool to simulate fluid models of high-speed networks. The aim of such a simulator is to evaluate measures which cannot be obtained through standard tools in reasonable time or through analytical approach.

G.C.Avontuur and K.van der Werff

G.C.Avontuur and K.van der Werff (2017) has presented a performance evaluation of the Systems reliability analysis of mechanical and hydraulic drive systems, an Academic Journal of Manufacturing Engineering. Here a method based on finite element equations, which describe motion and equilibrium between internal and external loads for structures and mechanisms. The theory has been extended to describe the simple flow of fluids.

Thomas Schwarzgruber and Passenbrunner

Schwarzgruber and Passenbrunner (2014) presented their work on Control

design for a multi input single output hydraulic cylinder. Their work discusses several approaches to the multi input, single output problem in hydraulic control by means of cascade methods. The multi input problem arises from the fact, that the actuator force in a hydraulic cylinder is the result of a combination of two pressure states, which is not uniquely defined.

Ali M. Rushdi and Husain A. Al-Yahya

Ali and Husain (2000) presented an approach for reducing the equations that are Variable entered in Karnaugh Map. It procedures for obtaining the irredundant disjunctive forms of a switching function from its complete sum. The paper advocates the use of the variable-entered Karnaugh map to solve the classical problem of designing hydraulic circuit by intuitive methods.

CHAPTER 3

METHODOLOGY

3.1 HYDRAULIC SYSTEM IMPLEMENTATION

Hydraulic systems are power-transmitting assemblies employing pressurized liquid as a fluid for transmitting energy from an energy-generating source to an energy-using point to accomplish useful work. Fig 3.1 shows a simple circuit of a hydraulic system with basic components. Functions of the components shown in Fig. 3.1 are as follows:

1. The hydraulic actuator is a device used to convert the fluid power into mechanical power to do useful work. The actuator may be of the linear type (e.g., hydraulic cylinder) or rotary type (e.g., hydraulic motor) to provide linear or rotary motion, respectively.
2. The hydraulic pump is used to force the fluid from the reservoir to rest of the hydraulic circuit by converting mechanical energy into hydraulic energy.
3. Valves are used to control the direction, pressure and flow rate of a fluid flowing through the circuit.
4. External power supply (motor) is required to drive the pump.
5. Reservoir is used to hold the hydraulic liquid, usually hydraulic oil.
6. Piping system carries the hydraulic oil from one place to another.
7. Filters are used to remove any foreign particles so as keep the fluid system clean and efficient, as well as avoid damage to the actuator and valves.
8. Pressure regulator regulates (i.e., maintains) the required level of pressure in the hydraulic fluid.

The piping shown in Fig. 3.1 is of closed-loop type with fluid transferred from the storage tank to one side of the piston and returned back from the other side of the piston to the tank. Fluid is drawn from the tank by a pump that produces

fluid flow at the required level of pressure. If the fluid pressure exceeds the required level, then the excess fluid returns back to the reservoir and remains there until the pressure acquires the required level.

Cylinder movement is controlled by a three-position change over a control valve.

1. When the piston of the valve is changed to upper position, the pipe pressure line is connected to port A and thus the load is raised.
2. When the position of the valve is changed to lower position, the pipe pressure line is connected to port B and thus the load is lowered.
3. When the valve is at center position, it locks the fluid into the cylinder (thereby holding it in position) and dead-ends the fluid line (causing all the pump output fluid to return to tank via the pressure relief).

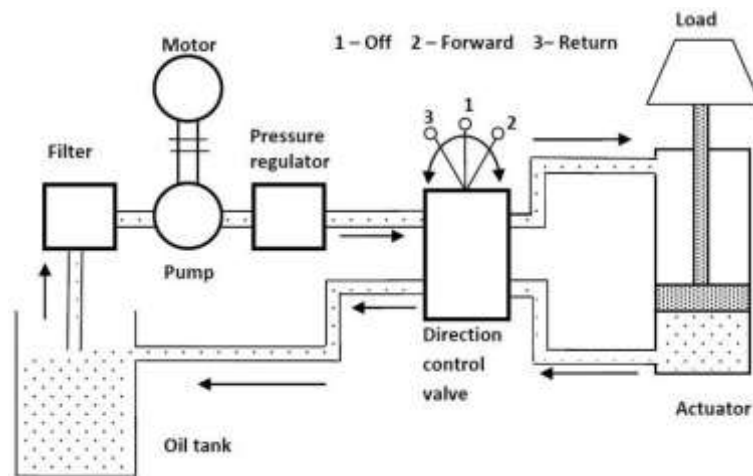
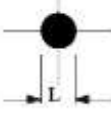
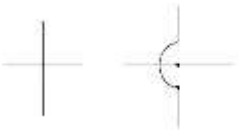
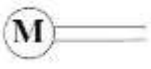
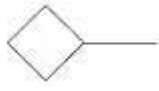
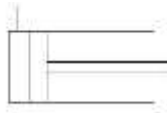
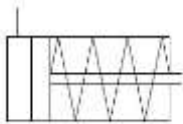


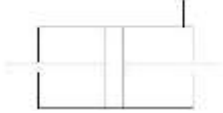
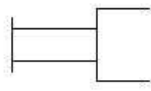
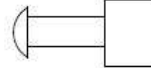
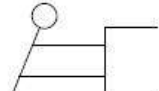
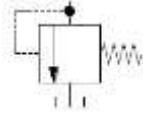
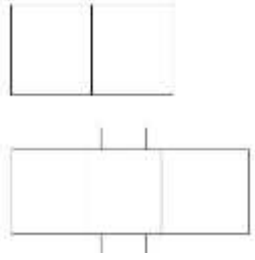
Fig 3.1 Components of a hydraulic system

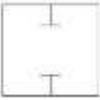
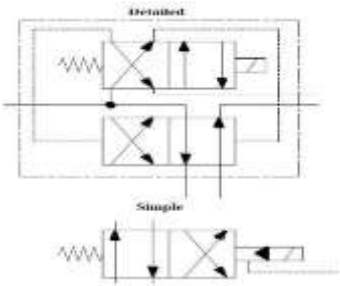
3.2 USAGE OF SYMBOLS IN IMPLEMENTATION

Hydraulic symbols provide a clear representation of the function of each hydraulic component. Laying each symbol out on the page in the same sequence the components are used in the circuit allows people to understand the complete function of the hydraulic equipment. Table 3.1 shows the International Organization for Standardization (ISO) symbols used in the implementation of hydraulics circuits.

Table 3.1 ISO symbols for fluid power elements

Component	Standard Symbol
Hydraulic	
Continuous flow	
Connected	
Not connected	
Spring	
Electric motor	
Filter or strainer	
Reservoir	
Pressure gauge	
Fixed-capacity hydraulic pump	
Fixed-capacity hydraulic motor	
Cylinder, single-acting	
Cylinder, returned by spring	

Cylinder, double-acting with single piston rod	
Cylinder, double-acting with double piston rod	
Control by muscular	
Control by push button	
Control by lever	
Control by solenoid	
Control by application of pressure	
Pressure relief valve	
These squares indicate direction control valves, having as many distinct positions as there are squares. The pipe connections are represented as connected to the box, representing the unoperated condition. For operation, boxes are imagined to be displaced, so that pipe connections correspond with ports of the box in question	
Two closed ports	
One-flow path	

Two closed ports	
Directional control valve (DCV) single-solenoid controlled, pilot-operated in both directions, spring return, four ports, two distinct position	

The hydraulic system discussed in Fig 3.2 is the symbolic representation which can be broken down into four main divisions that are analogous to the four main divisions in an electrical system.

1. The power device parallels the electrical generating station.
2. The control valves parallel the switches, resistors, timers, pressure switches, relays, etc.
3. The lines in which the fluid power flows parallel the electrical lines.
4. The fluid power motor (whether it is a rotating or a non-rotating cylinder or a fluid power motor) parallels the solenoids and electrical motors.

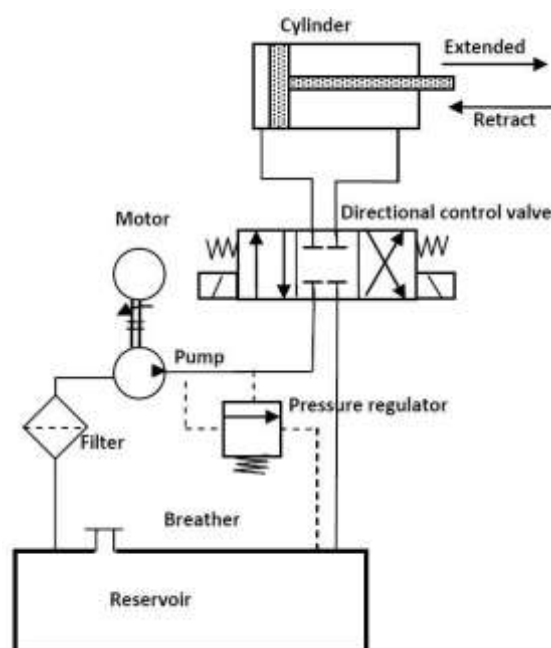


Fig 3.2 Components of a hydraulic system (shown using ISO symbols)

3.3 WORKING PRINCIPLE

Pascal's law and Bernoulli's principle operate at the heart of all fluid power applications and are used for analytical purposes. The hydraulic system works on the principle of Pascal's law which states that 'the external static pressure applied on a confined liquid is distributed or transmitted evenly throughout the liquid in all directions'. In simple words the pressure in an enclosed fluid is uniform in all the directions.

The formula for Pascal's principle is given as follows:

$$F = PA$$

-Where F is the force applied, P is the pressure transmitted, and A is the cross-sectional area.

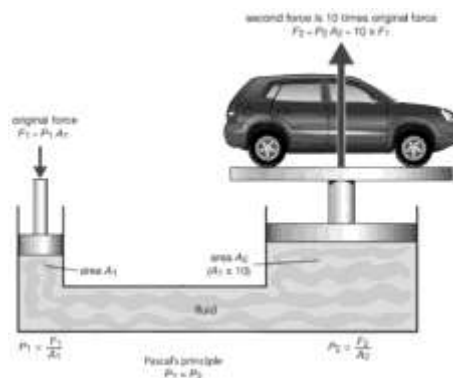


Fig 3.3 Hydraulic lift explaining Pascal's principle (<https://byjus.com>)

Bernoulli's principle deals with the law of conservation of energy for a fluid flowing in a pipeline, which states that 'an increase in the speed of the fluid occurs simultaneously with decrease in pressure or decrease in fluid potential energy'. Bernoulli's equation formula is a relation between pressure, kinetic energy, and gravitational potential energy of a fluid in a container. The formula for Bernoulli's principle is given as follows:

$$p + \frac{1}{2} \rho v^2 + \rho gh = \text{constant}$$

-Where p is the pressure exerted by the fluid, v is the velocity of the fluid, ρ is the density of the fluid and h is the height of the container.

Bernoulli's equation gives great insight into the balance between pressure, velocity and elevation.

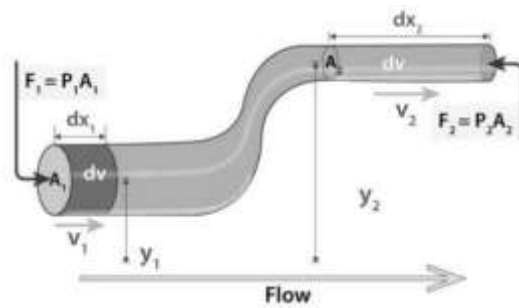


Fig 3.4 Pipe with varying diameter and height explaining Bernoulli's principle (<https://byjus.com>)

CHAPTER 4

STRUCTURE-DESIGN AND IMPLEMENTATION

The design of our intendant sequence (A+B+B-A-) can be made by two commonly used methods, the proposed circuit design can be implemented and simulated in FluidSIM-H software.

1. Cascade Method
2. KV map Method

4.1 CASCADE METHOD

Cascade Method involves dividing the sequence into groups with each group's manifold (power or main pressure line) being supplied with pneumatic power (pressure) one at a time and in sequence. Motion within each group is powered by its own group manifold.

Following are the design steps in cascade method:

1. Assign letters A, B, ... for each cylinders in the sequence.
2. Divide the sequence into groups so that no letter is repeated within any group.
3. For each cylinder assign a 4/2 or 5/2 DCV with double pilot lines and two spring return limit switch valves to indicate the end of strokes.
4. Assign supply lines S1, S2, ... which is equal to number of groups in the sequence.
5. Assign one or more group valve or cascade valve (DCV) to control fluid pressure to group manifold lines or supply lines so that only one group manifold line is pressurized at any time and in sequence.
6. Connect the input of limit valve at each stroke to its one group manifold.
7. For the first letter in each group, connect the corresponding DCV input to its one group manifold.
8. For the last letter in each group, connect the output of the corresponding limit valve to switch to the next group manifold.

9. For all other letters, connect the output of the corresponding limit valve to control valve input corresponding to the next letter in the sequence.

The circuit diagram according to cascade method is as shown in the Fig 4.1.

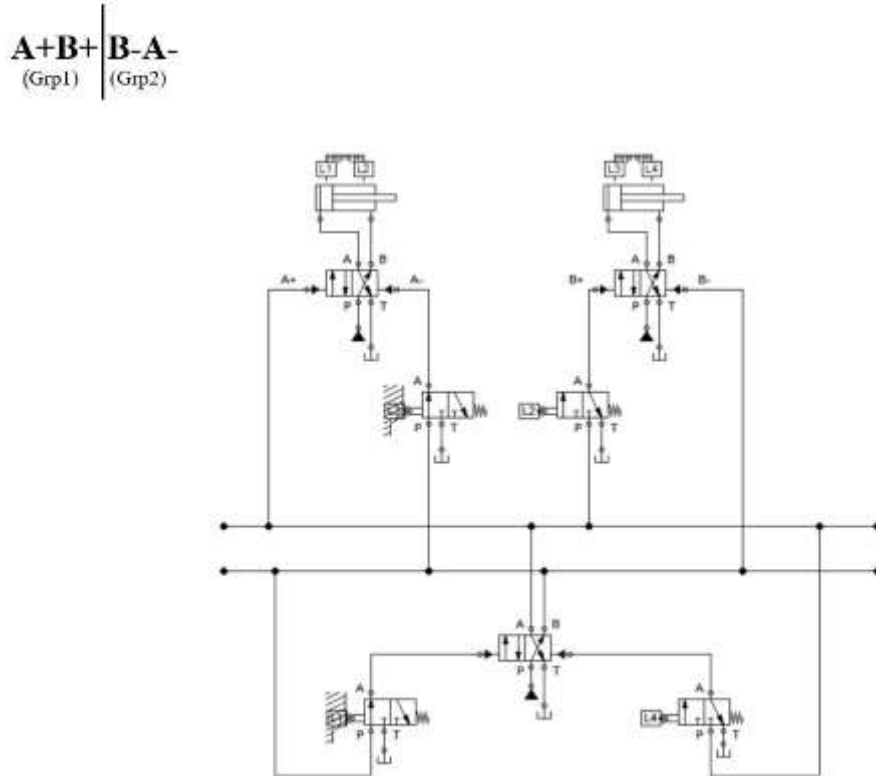


Fig 4.1 Design of A+B+B-A- using Cascade Method

4.2 KV MAP METHOD

Karnaugh Veitch (KV) map minimizes logical conditions used for ON/OFF control in hydraulic circuits. The methodology ensures not only the desired sequential cycle, as well as the minimization of the variables necessary for control commands.

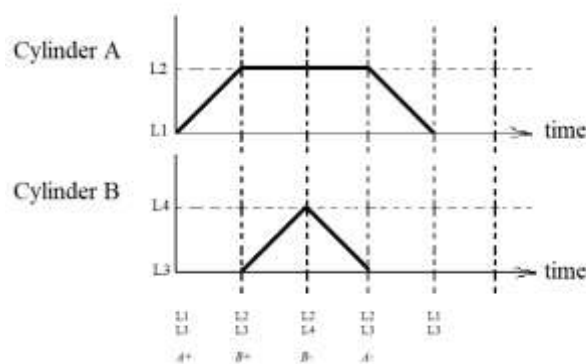
The KV map is a method of simplifying Boolean algebra expressions. It reduces the need for extensive calculations by taking advantage of humans' pattern-recognition capability and involves simple logical calculation. The previous method is a time consuming process and requires more human skills, thus its error prone. KV map method is a graphical method which is easy for the simplification of Boolean functions.

Following are the design steps in KV map method:

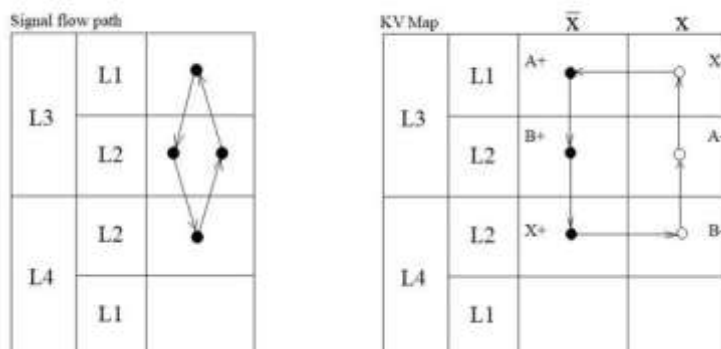
1. Make the statement of the sequence and draw its traverse time diagram including all the cylinders in the sequence.
2. Prepare a signal flow diagram and plot the signal flow path, then prepare KV map for every memory (DCV) and for every auxiliary memory appearing in the signal flow diagram.
3. Extend the presence of memory designations by filling and forming loops, then extract minimized memory set and reset the equations.
4. Construct the hydraulic circuit as per the minimized set of equations.

The graphical part of KV map is as in the Fig 4.2

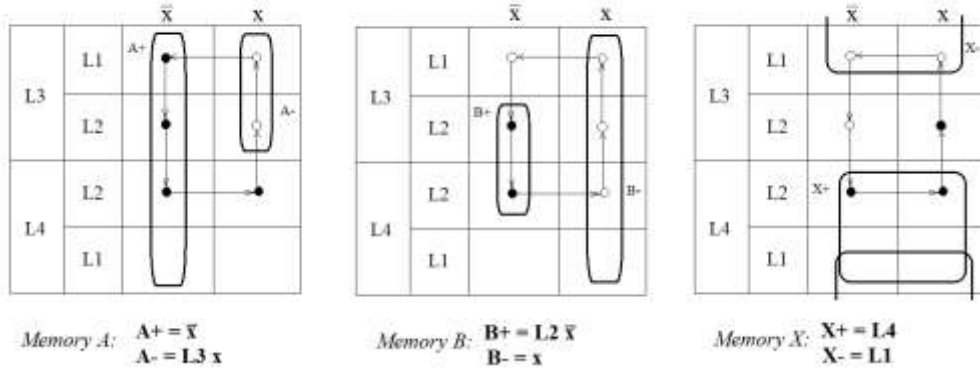
Step 1 : Traverse Time Diagram



Step 2 : Signal Flow Path and KV Map



Step 3 : Minimization After Looping



Step 4 : Circuit Design Based on the equations

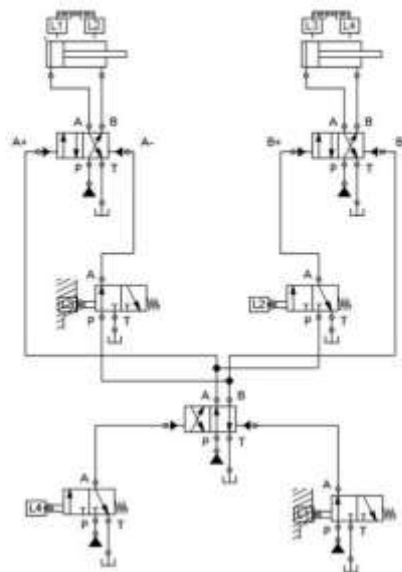


Fig 4.2 Design of A+B+B-A- using KV map Method

4.3 MATERIAL STUDY

Without the component analysis and material study a project under Mechanical Engineering is incomplete. So as per Mechanical Engineering is concerned, the material science technology and manufacturing process part is analyzed for the machine tool operation ‘Automatic Clamping and Drilling’. Normally the drill bit used can be High-speed steel (HSS). It is a popular material good for drilling into soft steels as well as wood and plastic. Typical component study and its description is as in the Table 4.1.

Table 4.1 Component and Material study

Component	Material used	Description
Hydraulic Cylinder	Stainless Steel Type 304	Type 304 Stainless is one of the most widely used materials in engineering. They are not only resistant to corrosion but is also easily available and environmentally friendly. They are durable in acidic environments and does not wear easily.
Hydraulic Piston	Cast aluminum alloy	Pistons made of cast aluminum alloy are excellent and lightweight thermal conductivity. Thermal conductivity is the ability of a material to conduct and transfer heat. Aluminum expands when heated, and proper clearance must be provided to maintain free piston movement in the cylinder bore.
Hydraulic Oil	Castrol HYPIN HEAVY DUTY 32 Hydraulic Oil	Castrol Hyspin Heavy-duty hydraulic oil range is based upon highly refined mineral oil enhanced with a stabilized zinc additive system.
DCV	Monel	Monel is a nickel-copper alloy which offers superior corrosion resistance. It is typically used for cladding of valve trim parts.
Hydraulic Tank	Aluminized steel	The use of aluminized steel (sheet steel hot dip coated on both sides with an aluminum-silicon alloy) solves the contamination problems inherent in uncoated steels.

Hydraulic Pump	Plastics, synthetic rubbers, and steel alloys	Plastics, synthetic rubbers, and steel alloys are used as a material for fabrication. The fabricating materials are selected depending on the exposed temperature and pressure of these pumps. Titanium alloys and polymers are high strength materials used for high-pressure applications.
Hydraulic Pipes and Working lines	Steel pipes, copper pipes, stainless steel pipes, and are used in piping for hydraulic equipment. Copper pipes are not used in common practice because it accelerates oxidation of petroleum-based hydraulic fluids.	Hydraulic flow lines are depicted as single straight lines for both pressure and return lines. Although there is no specific hose that is designed to be a return line, it is still a vital piece in a system. Return lines are responsible for bringing fluids back to the pump where the pressure decreases.

4.4 APPLICATION

Hydraulic circuits are used in industrial and mobile applications to transmit power from a prime mover to operate machine parts or vehicles. Stationary applications include tool machines like presses.

Hydraulic presses are commonly used for forging, clinching, molding, blanking, punching, deep drawing, and metal forming operations. Hydraulic presses are also used for stretch forming, rubber pad forming, and powder compacting.

Hydraulic Lifts and power packs are another common application of this field. Fig 4.3 to 4.5 shows the sketches of these applications.



Fig 4.3 Hydraulic Lift



Fig 4.4 Hydraulic Power Pack

Application of the proposed sequence A+B+B-A-

1. Automatic clamping and drilling operation
2. Automatic clamping and stamping operation

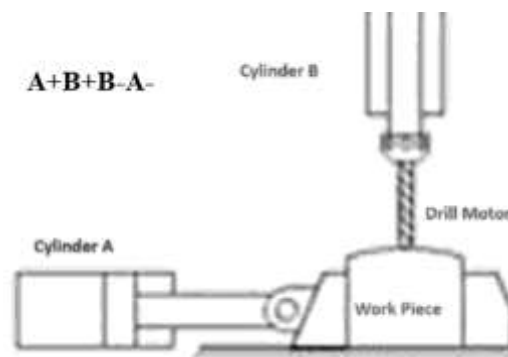


Fig 4.5 Automatic clamping and drilling operation

CHAPTER 5

RESULTS AND DISCUSSION

The design of the proposed sequence A+B+B-A- was made. It was simulated in FluidSIM-H using cascade and KV map method, and obtained the results successfully. A comparison chart of cascade and KV are given in the Table 5.1. Through this mini project we had recognized the general application of sequential hydraulic circuits. We also studied the working principle of the system and its component analysis.

Table 5.1 Comparison of Cascade method and KV map method

Sl. No.	Cascade Method	KV Map Method
1.	A type of classical or intuitive method.	A type of logical mapping method.
2.	More steps are involved here. Here mathematical computations are not used.	Lesser steps involved because of the use of mathematical computations.
3.	The concept of grouping is involved here, the number of groups is equal to the number of supply lines required.	The concept of grouping is not involved here.
4.	Traverse time diagram and signal flow path are not required.	They are required and these are the primary steps of KV map method.
5.	Circuits with N number of variables can be designed, but the task is very tedious as N increases.	It is not suitable when the number of variables involved exceeds four, Quine–McCluskey (QM) method can be used to design circuits for $N > 4$.
6.	Lesser optimized so higher cost to design the circuit when compared to KV.	More optimized so lesser cost to design the circuit when compared to cascade.

5.1 SIMULATION RESULTS

A screenshot of simulation from FluidSIM-H software is attached hereby. Fig 5.1 shows the circuit design by cascade method and its corresponding sequential operations. Similarly Fig 5.2 shows the circuit design by KV map method and its corresponding sequential operations.

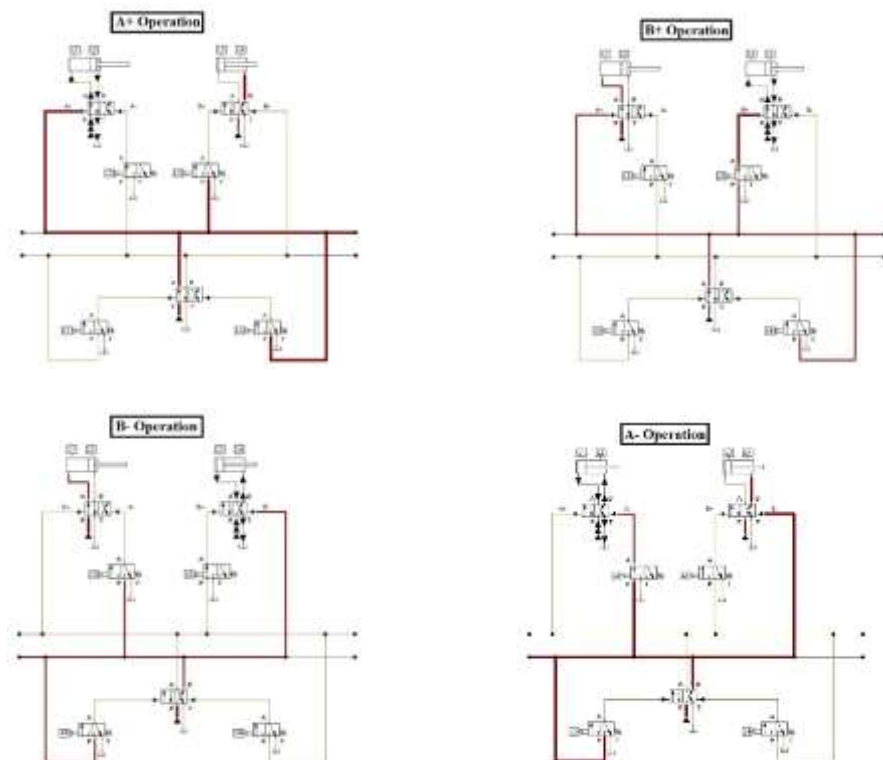
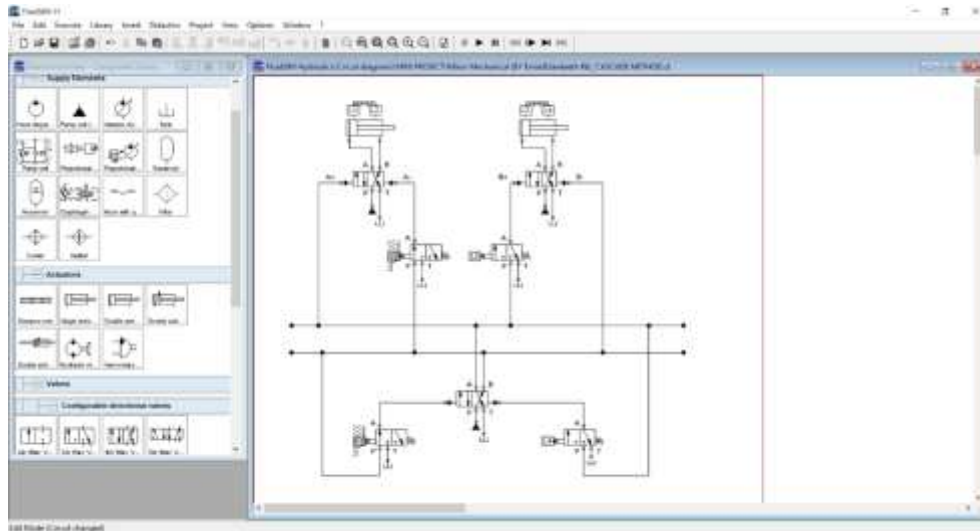


Fig 5.1 Simulation results of A+B+B-A- using Cascade Method

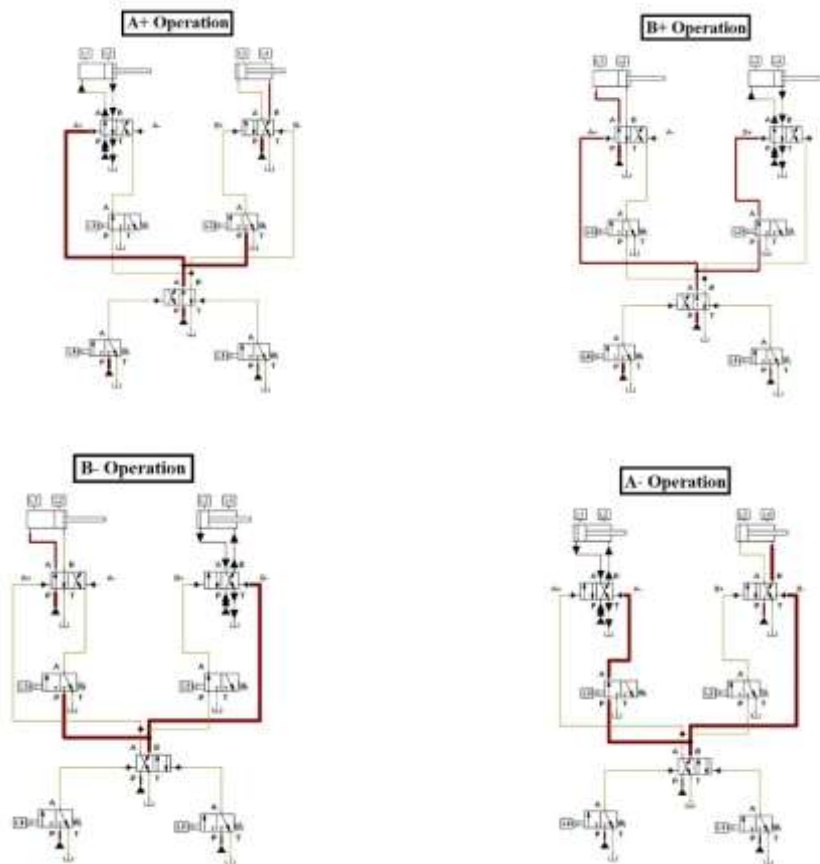
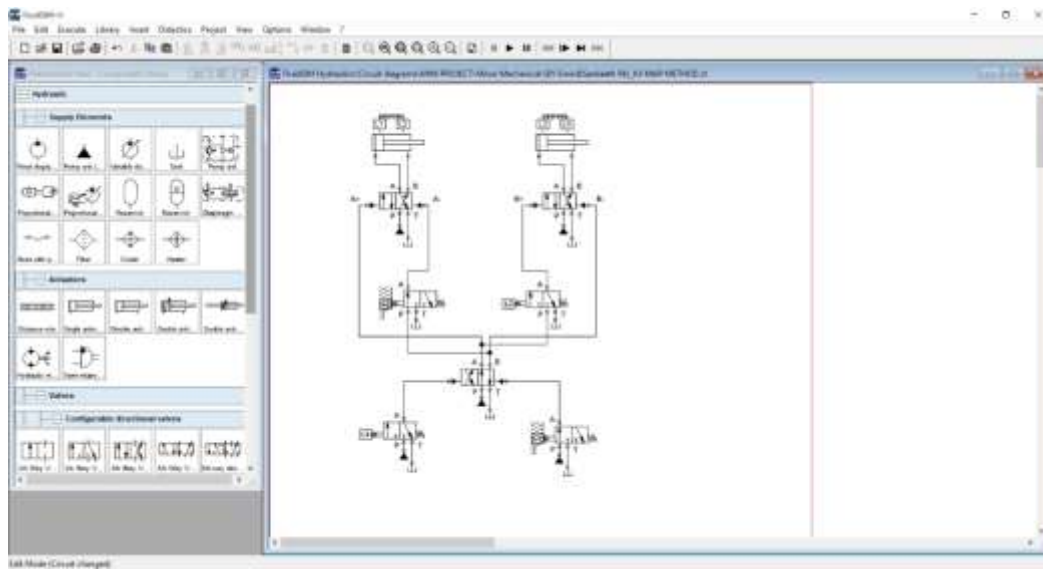


Fig 5.2 Simulation results of A+B+B-A- using KV map Method

CHAPTER 6

CONCLUSION

The automation market in India is estimated to be 1/10th that of China. If India has to become one of the leading economies in the world, based on manufacturing, it will have to attain higher technological standards and higher level of automation in manufacturing. In the past 30 years, fluid power technology rose as an important industry. With increasing emphasis on automation, quality control, safety and more efficient and green energy systems, fluid power technology should continue to expand in India.

The growth rate of this industry in India is typically about twice the growth of economy. The reasons for this are:

- As the economy grows, this industry grows
- There is a lot of automation and conversion into more sophisticated manufacturing methods which increases the rate
- One of the interesting things happening in this industry is that India is becoming an attractive destination for manufacturing and outsourcing of some of the products

We have designed our intended sequence (A+B+B-A-) using two methods, cascade and KV map in FluidSIM-H software and the simulation results was studied. We made a comparison chart of the two methods. We have also analyzed the materials used in the manufacturing of the proposed hydraulic systems. It have real life applications like automatic clamping and drilling operations.

We also correlate the theory subjects of minor studied in our previous semesters for developing this mini project, which are Material Science and Technology (MET 285), Manufacturing Technology (MET 286), Machine Tools Engineering (MET 385) and an NPTEL MOOC course on Advanced Machining Process (MEM 305).

CHAPTER 7

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